The Focus

One study of space in pieces, and the Hexagonality Law

DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. DICTO concluded that “everything is the same study now: all aspects of everything we have been looking at have come to a focus”. This paper sets out that focus: one map of how the threads of the recent papers meet, with the status of every link, and one new theorem that ties the flat stackings together. Stack equal balls in layers, each layer cubic or hexagonal by its neighbours, and share space out among the balls and the holes between them by weight, as real crystals do. **The Hexagonality Law:** in every stacking, each kind of piece totals exactly what its layers would hold in their own pure stackings, so the total is a straight line in the number of hexagonal layers, whatever their order. We prove it from the Swap Rule (#61) wherever a piece is fixed by its two neighbouring layers, show that it is equivalent to the Swap Rule there, and check it exactly in 1064 cases across pyrochlore, rock salt, cadmium iodide, zinc blende and the Laves phases, with no failure; in cadmium iodide, whose filled and empty layers alternate, it holds for each kind of layer separately. It holds even for the one Laves piece that reads three layers, where it is not yet proved. In the physics of stacking variants, straight-line laws in the hexagonality have been measured for sixty years; this is an exact geometric law of that form.

1. The focus

Over the last papers the threads of this work converged: the rule of halves for stackings of balls (#57), its pyrochlore and crystal forms (#60, #64), the Swap Rule (#61), the White Queen 600-cell and its Cleo shells (#62, #63), and the Red Queen 120-cell with Aristotle's five pyramids at the cores of the four-dimensional solids (#65). DICTO concluded that “everything is the same study now: all aspects of everything we have been looking at have come to a focus”. The common thread is that space, in the flat stackings and in the four-dimensional solids alike, is shared out into a few kinds of pieces built from the tetrahedron and the octahedron, whole or in halves, and that when a stacking or a view changes, pieces change by swapping. This paper does two things. It draws the map (§5), with the status of every link, and it proves the theorem that makes “change by swapping” exact for the stackings: the Hexagonality Law (§3).

2. Stackings, layers and pieces

Close-packed layers of equal balls sit in three positions A, B, C, and each new layer sits on the one below in one of the two other positions, so a stacking is a word such as ABC (cubic, 3C), AB (hexagonal, 2H) or ABCB (4H). Each layer is *cubic* (c) when its two neighbours differ and *hexagonal* (h) when they are the same: Jagodzinski's notation. In a stacking with period n, n_c layers are cubic and n_h hexagonal; the hexagonality is n_h/n. Every gap between two layers holds a ball site at its foot, two tetrahedral sites and one octahedral site. A crystal family fills

some of these sites with atoms, and space is shared out among all sites by weight, in a power diagram, as in #57 and #64: atoms of one kind have weight λ (zinc blende's second atom λ + μ), empty holes weight 0. Each site's cell is a *piece*, and its *role* is its kind of site (ball, octahedral, one of the tetrahedral sites; for the Laves phases the six sites of their net, #64). Everything is exact in rational arithmetic, with volumes as polynomials in λ.

3. The Hexagonality Law

> **Theorem (Hexagonality Law).** Let a role have, in the gap between layers of types x (below) and y (above), a piece of volume F(x, y), the same in every stacking. Then the following are equivalent:
> (i) the Swap Rule F(c, h) + F(h, c) = F(c, c) + F(h, h);
> (ii) in every stacking, the total volume of the role over one period is $T = n_c \cdot F(c, c) + n_h \cdot F(h, h)$.
> So each layer holds, in every stacking, exactly what it holds in the pure stacking of its own type, and the total per layer is the straight line $F(c, c) + (n_h/n) \cdot (F(h, h) - F(c, c))$ in the hexagonality.

Proof. (i) ⇒ (ii). Put g(c) = 0, g(h) = F(c, h) − F(c, c) and f(x) = F(x, c). Then F(x, y) = f(x) + g(y) for all four pairs: for (x, c) by definition, for (c, h) by definition, and for (h, h) by the Swap Rule, since f(h) + g(h) = F(h, c) + F(c, h) − F(c, c) = F(h, h). Summing over the n gaps of one period, gap k between layers k and k + 1, gives $T = \Sigma_k f(t_k) + \Sigma_k g(t_{k+1}) = \Sigma_k (f + g)(t_k)$, because the period closes up. As (f + g)(c) = F(c, c) and (f + g)(h) = F(h, h), $T = n_c F(c, c) + n_h F(h, h)$. (ii) ⇒ (i). The stacking ABCB (4H) has layer types alternating h, c, so its four gaps are (h, c), (c, h), (h, c), (c, h) and T = 2F(h, c) + 2F(c, h); (ii) gives T = 2F(c, c) + 2F(h, h). ∎

Two kinds of layer. In cadmium chloride and cadmium iodide the octahedral sites are filled in every other gap only, so the layers come in two kinds, alternating. The same proof, with f and g allowed to depend on the kind, gives $T = f(c)n_c' + f(h)n_h' + g(c)n_c'' + g(h)n_h''$, where ′ and ″ count the layers of each kind: the law holds for the hexagonality of each kind of layer separately. A single hexagonality is not enough there: it fails exactly for the stackings ABCB and ABCBAB, the two in which the two kinds of layer have different hexagonalities.

Exact check. For every family of the crystal runs of #60 and #64, every close-packed stacking up to six layers (AB, ABC, ABCB, ABACB, ABCBAB, ABCACB, up to symmetry), every window of the weight λ and every role, we took the total volume over one period from the exact weight walks and tested it against $n_c \cdot T(ABC)/3 + n_h \cdot T(AB)/2$, as an identity of polynomials in λ.

Family	Exact tests	Linear in the hexagonality
Pyrochlore (#60)	96	96
Rock salt / nickel arsenide (#64)	80	80
Zinc blende / wurtzite, μ = ±1/20, ±1/10 (#64)	688	688
Laves phases (#64)	120	120
Cadmium chloride / iodide (#64), by kind of layer	80	80 (four unknowns from six stackings, consistent)
Total	**1064**	**1064**

(In cadmium iodide a single hexagonality was also tested, in 320 cases: 74 fail, all in ABCB and ABCBAB, as the two-kinds statement predicts.) The theorem's hypothesis, that a piece is fixed by its two neighbouring layers, was checked in #64 for every role except the Laves foot atom above λ = 17/96, which reads three layer types. For that role the law holds in every test above, but the proof does not apply: it is an open problem whether it holds for every stacking.

4. Physics

In the physics of stacking variants (polytypes), a property of a crystal is written as a sum over its layers, with terms for each layer alone and for pairs of layers further apart: the axial next-nearest-neighbour Ising (ANNNI) picture of Heine and coworkers for silicon carbide (Cheng, Needs and Heine, 1988). In that language the Hexagonality Law says that for the volume of every piece the pair terms are exactly zero: each layer contributes alone. Straight-line laws in the hexagonality are classical in measurement. The indirect band gap of the silicon carbide polytypes rises with the hexagonality, roughly in a straight line (Choyke, Hamilton and Patrick, 1964), and basal lattice constants of metallic polytypes correlate linearly with it; later work finds these laws approximate. The Hexagonality Law is an exact geometric law of the same form, for the space each atom and hole occupies. We make no claim that it explains the measured laws; it is a candidate for an exact part of them. Two further echoes, of form only: causal dynamical triangulations build spacetime in slabs between slices, from simplices typed by how they meet the slice above and the slice below, as our pieces are typed by the layers above and below their gap; and Regge calculus has used the 600-cell as a model universe (De Felice and Fabri, 2001; Tsuda and Fujiwara, 2020).

5. The map

Each thread, its link to the focus, and its status. *Proved*: an exact proof in a published paper or this one. *Proved (exact)*: exact computation in a committed study, not yet in a paper. *To test*, *stage 0*: planned. *Likeness only*: a visual likeness, not a theorem.

Thread	Link to the focus	Status
Atoms — the octet pair	the regular tetrahedron and octahedron: a tetrahedron is an octahedron with four small tetrahedra on alternate faces; both together fill space (the octet)	classical
Rule of halves (#57)	every stacking of balls shares space into pieces borrowed whole or in half from the pure cubic and hexagonal stackings	proved
Pyrochlore halves (#60)	corner-sharing tetrahedra on every stacking keep the rule of halves	proved
Swap Rule (#61)	squashed stackings: the two mixed pieces add up to the two pure ones; true close packing is the border where the tetrahedral pieces become Janus pieces (Ganesha)	proved
Swap Rule in crystals (#64)	rock salt, cadmium iodide, zinc blende and the Laves phases: the Swap Rule in all 299 testable cases	proved
Hexagonality Law (this paper)	in every family, each kind of piece totals exactly what its layers hold in their own pure stackings: a straight line in the hexagonality	proved; exact computation for one Laves role
White Queen 600-cell (#62, #63)	62 slice patterns in the four most symmetric directions, exact; 12 all triangles; four shadows each filled once by front and once by back cells	proved
Cleo shells (#63)	round one vertex the 600 cells stand in 15 shells, each one kind of piece, round a core of 20 Cleo-Icos; the counts read the same backwards	proved
Five pyramids (#44, #65)	Aristotle's five pyramids are the cores of the regular four-dimensional solids seen in perspective, with exactly his proportions	proved
Red Queen 120-cell (#65)	built exactly from the White Queen; 124 slice patterns; her shadows are polar to the White Queen's middle slices	proved
Boole Stott 1900	the cell-first sections of both queens at their corner layers are Alicia Boole Stott's; the other directions, the gaps and the shadows are ours	read; credited
DICTO's radial cell projection	builds the 600-cell cell by cell, folding at the angle with cosine (1 + 3√5)/8; its 16 steps from a cell are the adjacency shells of the exact model	proved (exact)
Six turnings	the six turnings between the main views; 14 swap moments; cells swap front and back in pairs	proved (exact)
Shadow-17 (#62)	the 17-corner edge-first slice is two J11 glued on their pentagons; J11, the diminished icosahedron, is the piece DICTO's work began with	proved
Kyushindo Cores (#43)	golden nested icosahedral shells, echoed by the slices and shadows of the 600-cell	to test
Gnomon ladders	every turning angle has cosine √(a + b√5), so it can be built from gnomons	stage 0
Scutoids	Janus and Ganesha pieces, and the pure pieces, are scutoids	likeness only

Named in the two queens, for later papers (DICTO's names): in the White Queen's atlas Humpty Dumpty, Shadow-17, the Lion and the Unicorn, Tweedledum and Tweedledee, the Jabberwock, the Jubjub and the Bandersnatch; in the Red Queen's the Red King, the Walrus, the Red and White Knights, Haigha and Hatta.

6. Conclusion

Space, in flat stackings and in the four-dimensional 600-cell alike, is shared out into a few kinds of pieces built from the tetrahedron and the octahedron, whole or in halves, and when a view or a stacking changes, the pieces change by swapping, never by appearing from nothing.

For the stackings the last clause is now a theorem: the Swap Rule is exactly the statement that every layer keeps, in every stacking, what it holds in its own pure stacking, so every kind of piece totals a straight line in the hexagonality. It holds in every crystal family we have computed, and in cadmium iodide for each kind of layer.

7. Prior art and credit

Jagodzinski's c/h (h/k) notation, the hexagonality and the polytypes are classical (Verma and Krishna, *Polymorphism and Polytypism in Crystals*, 1966). The ANNNI description of polytype energies is Heine's (Cheng, Needs and Heine, 1988). Linear trends of band gaps and lattice constants in the hexagonality are measured results (Choyke, Hamilton and Patrick, 1964), and an Ising-type additivity model has been used to estimate the band gaps of long polytypes (arXiv 2306.17756). We found no statement that the volumes of the weighted pieces of a close-packed stacking are exactly linear in the hexagonality, or that this is equivalent to the Swap Rule. The rule of halves and the Swap Rule are #57, #61 and #64.

> **Sources checked** (28 October 2026). *Reference library* (ask-library, five searches: polytype stacking energy and ANNNI; loop quantum gravity polyhedra; causal dynamical triangulations; Voronoi volumes in polytypes; Regge calculus): E. Rodriguez-Horta et al., close packed structure with finite range interaction (2017); R. Ganesh and A. K. Nasr, correlations in randomly stacked solids (2023); E. Bianchi, P. Doná and S. Speziale (2011); C. Rovelli and L. Smolin (1995); C. Rovelli (1998); S. Surya (2019); R. Tsuda and T. Fujiwara (2020); A. De Felice and E. Fabri (2001); V. A. Blatov (2004), catalogue entries and first pages. *Web* (search summaries and abstracts, not full texts): arXiv cond-mat/9709191 (SiC polytype energies, citing Cheng, Needs and Heine 1988); arXiv 2306.17756 (Ising-type additivity model for polytype band gaps); University of Pittsburgh, tribute to W. J. Choyke (Choyke, Hamilton and Patrick, Phys. Rev. 133, A1163, 1964); arXiv 1404.3588 (Matsushita and Oshiyama, on the linearity of the band gap in the hexagonality); Phys. Rev. B 49, 7564 (SiC band structures); ECS meeting 236 abstract (lattice constants of metallic polytypes against hexagonality); arXiv 2401.09399, 1111.6938, 1704.00577 (causal dynamical triangulations); arXiv 1508.05937, 2303.01761 (Voronoi topology of fcc and hcp). The 1988 and 1964 papers themselves were not read.

> **Data and code.** Study 2026-10-28-halves-physics (scripts/hexlaw.py, hexlaw_cdi2.py; out/hexlaw.json, hexlaw_cdi2.json), reading the exact weight walks of study 2026-10-27-splash-rules (run2/raw), attached to this record. References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; The Swap Rule (#61), 10.5281/zenodo.23682596; The Swap Rule in Crystals (#64), 10.5281/zenodo.23709753; The White Queen 600-cell (#63), 10.5281/zenodo.23690856; The Red Queen 120-cell (#65), 10.5281/zenodo.23716140.